

Decision Analysis

- ❑ The successes and failures of any business depends on the decisions taken by decision makers.
- ❑ Decision analysis is a systematic approach to the study of decision making.
- ❑ In this we study decision models that help decision maker to take best possible decisions.

Decision Analysis

The Five Steps in Decision Analysis

1. Clearly define the problem at hand.

E.g., John Thompson is the founder and president of Thompson Lumber Company. John identifies his decision-making problem as whether to expand his business by manufacturing and marketing a new product, backyard storage sheds.

✓ Decision problem: whether to expand his business by manufacturing and marketing a new product, backyard storage sheds.

2. The second step is to generate the complete list of decision alternatives available to the decision maker.

✓ Decision alternative is defined as a course of action that is available to the decision maker.

In Thompson Lumber's case, let us assume that John decides that his alternatives are as follows: (1) build a large plant to manufacture the storage sheds, (2) build a small plant to manufacture the storage sheds, or (3) build no plant at all.

Decision Analysis

The Five Steps in Decision Analysis

3. The third step involves identifying all possible future states of nature/ market conditions or outcomes for each decision alternative.

In decision analysis, outcomes are also known as states of nature.

In Thompson Lumber's case, suppose John determines that all three of his decision alternatives have the same three possible outcomes:

- (1) Demand for the sheds will be high,
- (2) demand for the sheds will be moderate, or
- (3) demand for the sheds will be low.

Decision Analysis

The Five Steps in Decision Analysis

4. In the fourth step, we need to identify the output that will result (profit/loss(negative profit)) if we choose a specific decision alternative, under a particular market condition a specific business environment.

In decision analysis, we call these outputs/results (profit/loss(negative profit)) as *payoffs*, regardless of whether they denote profit or cost.

E. g., In Thompson Lumber's case, let us assume that John wants to use net profits to measure his payoffs. He has already evaluated the potential profits associated with the various combinations of alternatives and outcomes, as follows:

- a) If John decides to build a large plant, he thinks that with high demand for sheds, the result would be a net profit of \$200,000 to his firm. The net profit would, however, be only \$100,000 if demand were moderate. If demand were low, there would actually be a net loss of \$120,000.
- b) If he builds a small plant, the results would be a net profit of \$90,000 if there were high demand for sheds, a net profit of \$50,000 if there were moderate demand, and a net loss of \$20,000 if there were low demand.
- c) Finally, doing nothing would result in \$0 payoff in any demand scenario

Decision Analysis

The Five Steps in Decision Analysis

ALTERNATIVES	OUTCOMES		
	HIGH DEMAND	MODERATE DEMAND	LOW DEMAND
Build large plant	\$200,000	\$100,000	-\$120,000
Build small plant	\$ 90,000	\$ 50,000	-\$ 20,000
No plant	\$ 0	\$ 0	\$ 0

5. Select one of the decision analysis modeling techniques.
 - ✓ Apply the decision model to the data and make your decision.
 - ✓ Selection of the decision model based on decision-making environment

Decision Analysis

Decision-making environments

1. Decision making under certainty:

Decision maker exactly knows what will happen in the future.

2. Decision making under uncertainty:

Decision makers have no information at all about the various outcomes.

They do not know the likelihood (or probability) that a specific outcome will occur.

3. Decision making under risks:

They know the likelihood (or probability) that a specific outcome will occur.

Decision Analysis

Decision making under uncertainty

- ✓ If the payoffs represent profit (i.e., We prefer higher payoffs to smaller ones).
- ✓ If the payoffs represent costs (i.e., we prefer smaller payoffs to higher ones).
- ✓ To avoid this confusion, convert costs in a payoff table to profits by multiplying all cost values by -1 .
- ✓ And apply the criteria as discussed here for all problems

Decision Analysis

Decision making under uncertainty Model

- a. **Maximax** (computed directly from the decision (payoff) table)
- b. **Maximin** (computed directly from the decision (payoff) table)
- c. **Criterion of realism or Hurwicz** (computed directly from the decision (payoff) table)
- d. **Equally likely or Laplace** (computed directly from the decision (payoff) table)
- e. **Minimax regret** (computed from opportunity loss table)

Decision Analysis

Decision making under uncertainty

a). Maximax (computed directly from the decision (payoff) table)

- ✓ This model selects the decision alternative that maximizes the maximum payoff over all alternatives.
- ✓ First locate the maximum payoff for each alternative and then select the alternative with the highest value among these maximum payoffs.
- ✓ It is also called the Optimistic criterion.

ALTERNATIVES	OUTCOMES			MAXIMUM FOR ALTERNATIVE
	HIGH DEMAND	MODERATE DEMAND	LOW DEMAND	
Build large plant	\$200,000	\$100,000	-\$120,000	\$200,000 → Maximax
Build small plant	\$ 90,000	\$ 50,000	-\$ 20,000	\$ 90,000
No plant	\$ 0	\$ 0	\$ 0	\$ 0

Decision Analysis

Decision making under uncertainty

b). **Maximin** (computed directly from the decision (payoff) table)

- ✓ Finds the alternative that maximizes the minimum payoff over all decision alternatives.
- ✓ First locate the minimum payoff for each alternative and then select the alternative with the highest value among those minimum payoffs.
- ✓ It is also called the pessimistic criterion.

ALTERNATIVES	OUTCOMES			MINIMUM FOR ALTERNATIVE
	HIGH DEMAND	MODERATE DEMAND	LOW DEMAND	
Build large plant	\$200,000	\$100,000	−\$120,000	−\$120,000
Build small plant	\$ 90,000	\$ 50,000	−\$ 20,000	−\$ 20,000
No plant	\$ 0	\$ 0	\$ 0	\$ 0 → Maximin

Decision Analysis

Decision making under uncertainty

c). Criterion of realism or Hurwicz (computed directly from the decision (payoff) table)

- ✓ The criterion of realism uses the weighted average approach.
- ✓ In this criterion, we use a parameter called the **coefficient of realism** to measure the decision maker's level of optimism regarding the future.
- ✓ This coefficient, denoted by a , has a value between 0 and 1.

Decision Analysis

Decision making under uncertainty Model

c). **Criterion of realism or Hurwicz** (computed directly from the decision (payoff) table)

$$\text{Realism payoff for alternative} = \alpha \times (\text{Maximum payoff for alternative}) \\ + (1 - \alpha) \times (\text{Minimum payoff for alternative})$$

ALTERNATIVES	OUTCOMES			WT. AVG. ($\alpha = 0.45$) FOR ALTERNATIVE
	HIGH DEMAND	MODERATE DEMAND	LOW DEMAND	
Build large plant	\$200,000	\$100,000	-\$120,000	\$24,000
Build small plant	\$ 90,000	\$ 50,000	-\$ 20,000	\$29,500 → Realism
No plant	\$ 0	\$ 0	\$ 0	\$ 0

Decision Analysis

Decision making under uncertainty

Model

c). **Criterion of realism or Hurwicz** (computed directly from the decision (payoff) table)

Build large plant: $0.45 \times (200,000) + 0.55 \times (-120,000) = 90,000 - 66,000 = 24,000$

Build small plant: $0.45 \times (90,000) + 0.55 \times (-20,000) = 40,500 - 11,000 = 29,500$

Build No plant: $0.45 \times 0 + 0.55 \times 0 = 0$

Decision Analysis

Decision making under uncertainty

Model

d). **Equally likely or Laplace** (computed directly from the decision (payoff) table)

- ✓ The equally likely (or Laplace) criterion finds the decision alternative that has the highest average payoff
- ✓ We first calculate the average payoff for each alternative and then pick the alternative with the maximum average payoff.

Decision Analysis

Decision making under uncertainty Model

d). **Equally likely or Laplace** (computed directly from the decision (payoff) table)

$$\frac{200,000 + 100,000 - 120,000}{3} = 60,000$$

$$\frac{90,000 + 50,000 - 20,000}{3} = 40,000$$

$$\frac{0 + 0 + 0}{3} = 0$$

ALTERNATIVES	OUTCOMES			AVERAGE FOR ALTERNATIVE
	HIGH DEMAND	MODERATE DEMAND	LOW DEMAND	
Build large plant	\$200,000	\$100,000	-\$120,000	\$60,000 → Equally likely
Build small plant	\$ 90,000	\$ 50,000	-\$ 20,000	\$40,000
No plant	\$ 0	\$ 0	\$ 0	\$ 0

Decision Analysis

Decision making under uncertainty

Model

e). **Minimax regret** (computed from opportunity loss table)

- ✓ First develop the opportunity loss table from the payoff table.
- ✓ To do so, we subtract each payoff for a specific outcome from the best payoff for that outcome.
- ✓ For example, the best payoff with high demand is \$200,000 (corresponding to building a large plant). Hence, we subtract all payoffs for that outcome (i.e., in that column) from \$200,000.

ALTERNATIVES	OUTCOMES		
	HIGH DEMAND	MODERATE DEMAND	LOW DEMAND
Build large plant	$\$200,000 - \$200,000 = \$ 0$	$\$100,000 - \$100,000 = \$ 0$	$\$0 - (-\$120,000) = \$120,000$
Build small plant	$\$200,000 - \$ 90,000 = \$110,000$	$\$100,000 - \$ 50,000 = \$ 50,000$	$\$0 - (-\$ 20,000) = \$ 20,000$
No plant	$\$200,000 - \$ 0 = \$200,000$	$\$100,000 - \$ 0 = \$100,000$	$\$0 - \$ 0 = \$ 0$

Decision Analysis

Decision making under uncertainty

Model

e). **Minimax regret** (computed from opportunity loss table)

- ✓ Once the opportunity loss table has been constructed, we locate the maximum opportunity loss (regret) for each alternative.
- ✓ We then pick the alternative with the smallest value among these maximum regrets.

ALTERNATIVES	OUTCOMES			MAXIMUM FOR ALTERNATIVE
	HIGH DEMAND	MODERATE DEMAND	LOW DEMAND	
Build large plant	\$ 0	\$ 0	\$120,000	\$120,000
Build small plant	\$110,000	\$ 50,000	\$ 20,000	\$110,000 → Minimax
No plant	\$200,000	\$100,000	\$ 0	\$200,000

Steve's Mountain Bicycle Shop is considering three options for its facility next year. Steve can expand his current shop, move to a larger facility, or make no change. With a good market, the annual payoff would be \$76,000 if he expands, \$90,000 if he moves, and \$40,000 if he does nothing. With an average market, his payoffs will be \$30,000, \$41,000, and \$15,000, respectively. With a poor market, his payoff will be $-\$17,000$, $-\$28,000$, and \$4,000, respectively.

- (a) Which option should Steve choose if he uses the maximax criterion?
- (b) Which option should Steve choose if he uses the maximin criterion?
- (c) Which option should Steve choose if he uses the equally likely criterion?

In the environment of increased competition, a fitness club executive is considering the purchase of additional equipment. His alternatives, outcomes, and payoffs (profits) are shown in the following table:

EQUIPMENT	FAVORABLE MARKET	UNFAVORABLE MARKET
Acme	\$400,000	−\$175,000
Standard	\$280,000	−\$ 90,000
High Pro	\$ 95,000	−\$ 15,000

- (a) If the executive is an optimistic decision maker, which alternative will he likely choose?
- (b) If the executive is a pessimistic decision maker, which alternative will he likely choose?
- (c) Market research suggests the chance of a favorable market for fitness clubs is 76%. If the executive uses this analysis, which alternative will he likely choose?

Decision Analysis

Decision making under Risks Model

a). Expected Monetary Value

- ✓ Let us assume the probabilities of high demand, moderate demand, and low demand are 0.3, 0.5, and 0.2, respectively
- ✓ Given a decision table with payoffs and probability, we can determine the expected monetary value (EMV) for each alternative.
- ✓ Selecting the alternative with the highest expected monetary value.

Decision Analysis

Decision making under Risks Model

a). Expected Monetary Value

$$\begin{aligned} \text{EMV (Alternative } i) = & (\text{Payoff of first outcome}) \\ & \times (\text{Probability of first outcome}) \\ & + (\text{Payoff of second outcome}) \\ & \times (\text{Probability of second outcome}) \\ & + \dots + (\text{Payoff of last outcome}) \\ & \times (\text{Probability of last outcome}) \end{aligned}$$

The largest EMV, \$86,000, results from the first alternative, build a large plant.

ALTERNATIVES	OUTCOMES			EMV FOR ALTERNATIVE
	HIGH DEMAND	MODERATE DEMAND	LOW DEMAND	
Build large plant	\$200,000	\$100,000	-\$120,000	$\$200,000 \times 0.3 + \$100,000 \times 0.5 + (-\$120,000) \times 0.2 = \$86,000$
Build small plant	\$ 90,000	\$ 50,000	-\$ 20,000	$\$90,000 \times 0.3 + \$50,000 \times 0.5 + (-\$20,000) \times 0.2 = \$48,000$
No plant	\$ 0	\$ 0	\$ 0	$\$0 \times 0.3 + \$0 \times 0.5 + \$0 \times 0.2 = \$ 0$
Probabilities	0.3	0.5	0.2	

Decision Analysis

Decision making under Risks Model

b). Expected Opportunity Loss

Minimize expected opportunity loss (EOL).

$$\begin{aligned} \text{EOL (Alternative } i) = & (\text{Regret of first outcome}) \\ & \times (\text{Probability of first outcome}) \\ & + (\text{Regret of second outcome}) \\ & \times (\text{Probability of second outcome}) \\ & + \cdots + (\text{Regret of last outcome}) \\ & \times (\text{Probability of last outcome}) \end{aligned}$$

Decision Analysis

Decision making under Risks

Model

b). Expected Opportunity Loss

- ✓ Using minimum EOL as the decision criterion, the best decision would be the first alternative, build a large plant, with an EOL of \$24,000.

ALTERNATIVES	OUTCOMES			EOL FOR ALTERNATIVE
	HIGH DEMAND	MODERATE DEMAND	LOW DEMAND	
Build large plant	\$ 0	\$ 0	\$120,000	$\$0 \times 0.3 + \$0 \times 0.5 + \$120,000 \times 0.2 = \$ 24,000$
Build small plant	\$110,000	\$ 50,000	\$ 20,000	$\$110,000 \times 0.3 + \$50,000 \times 0.5 + \$20,000 \times 0.2 = \$ 62,000$
No plant	\$200,000	\$100,000	\$ 0	$\$200,000 \times 0.3 + \$100,000 \times 0.5 + \$0 \times 0.2 = \$110,000$
Probabilities	0.3	0.5	0.2	

Maria Rojas is considering the possibility of opening a small dress shop on Fairbanks Avenue, a few blocks from the university. She has located a good mall that attracts students. Her options are to open a small shop, a medium-sized shop, or no shop at all. The market for a dress shop can be good, average, or bad. The probabilities for these three possibilities are 0.2 for a good market, 0.5 for an average market, and 0.3 for a bad market. The net profit or loss for the medium-sized and small shops for the various market conditions are given in the following payoff table:

ALTERNATIVE	OUTCOMES		
	GOOD MARKET	AVERAGE MARKET	BAD MARKET
Small shop	\$ 75,000	\$25,000	-\$40,000
Medium-sized shop	\$100,000	\$35,000	-\$60,000
No shop	\$ 0	\$ 0	\$ 0
Probabilities	0.2	0.5	0.3

Building no shop at all yields no loss and no gain. What do you recommend?

Decision Analysis

Decision Trees

- ✓ Problem that presented in a decision table can also be illustrated graphically, called decision tree.
- ✓ A decision tree consists of nodes (or points) and arcs (or lines), just like a network.
- ✓ Decision trees contain decision nodes and outcome nodes.
- ✓ A decision tree usually begins with a decision node.

Decision Analysis

Decision Trees

- = A *decision* node. Arcs (lines) originating from a decision node denote all decision alternatives available to the decision maker at that node. Of these, the decision maker must select only one alternative.
- = An *outcome* node. Arcs (lines) originating from an outcome node denote all outcomes that could occur at that node. Of these, only one outcome will actually occur.

Decision Analysis

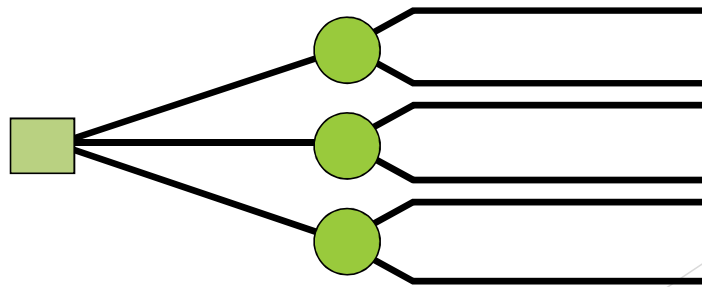
Decision Trees

Five steps to decision tree analysis

1. Define the problem
2. Structure or draw the decision tree
3. Assign probabilities
4. Estimate payoffs for each possible combination of alternatives
5. Solve the problem by computing expected monetary values (EMVs) for each alternative

Structure of Decision Trees

- ✓ Trees start from left to right
- ✓ Represent decisions and outcomes in sequential order
- ✓ Squares represent decision nodes
- ✓ Circles represent states of nature nodes (Outcome Node)
- ✓ Lines or branches connect the decisions nodes and the outcome node



Decision Analysis

Decision Tree

EXAMPLE PROBLEM: THOMPSON LUMBER COMPANY

Thompson Lumber Company is trying to decide whether to expand its product line by manufacturing and marketing a new product which is “backyard storage sheds.”

The courses of action that may be chosen include:

- (1) large plant to manufacture storage sheds,
- (2) small plant to manufacture storage sheds, or
- (3) build no plant at all.

THOMPSON LUMBER COMPANY

ALTERNATIVE	STATE OF NATURE	
	FAVORABLE MARKET (\$)	UNFAVORABLE MARKET (\$)
Construct a large plant	200,000	-180,000
Construct a small plant	100,000	-20,000
Do nothing	0	0
Probability	0.5	0.5

Thompson's Decision Tree

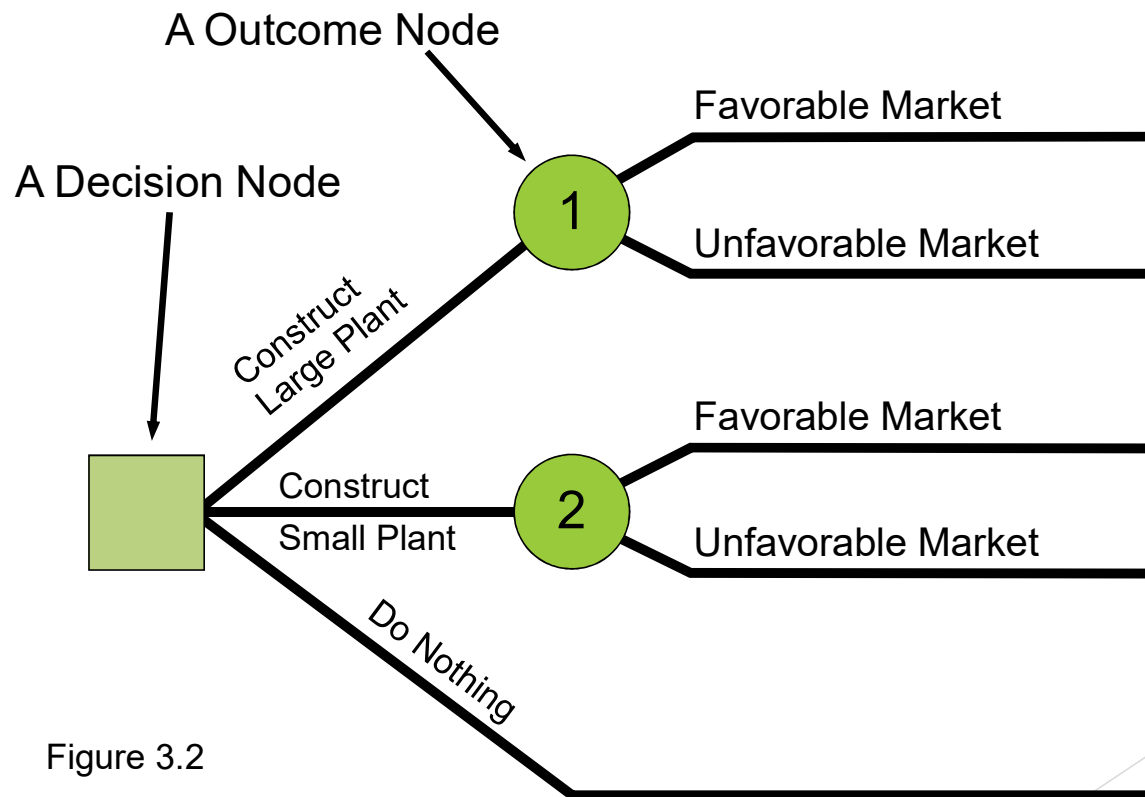


Figure 3.2

Thompson's Decision Tree

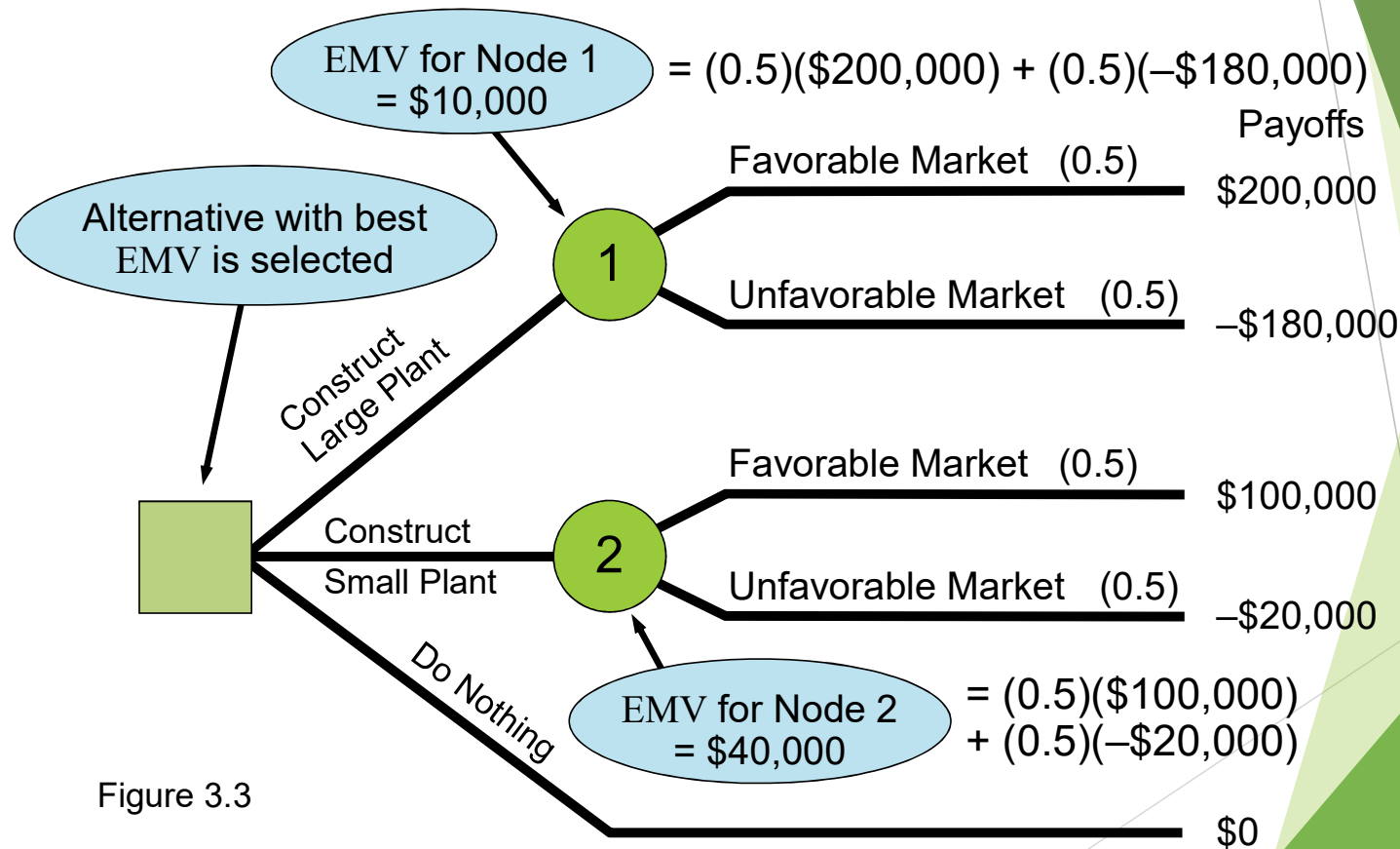


Figure 3.3

Decision Analysis

Decision Trees

Folding back of decision tree

- ✓ The process by which a decision tree is analyzed to identify the optimal decision is called folding back the decision tree.
- ✓ We start with the payoffs (i.e., the right extreme of the tree) and work our way back to the first decision node.
- ✓ The EMV is calculated at each outcome node.
- ✓ Using the rule, we now select the alternative with the highest EMV.
- ✓ The best alternative is selected at a decision node.

analyze his decision problem in an orderly, rational manner.

Example 10-17. An executive has to make a decision. He has four alternatives D_1 , D_2 , D_3 and D_4 . When the decision has been made events may lead such that any of the four results may occur. The results are R_1 , R_2 , R_3 and R_4 . Probabilities of occurrence of these results are as follows :

$$R_1 = 0.5, R_2 = 0.2, R_3 = 0.2, R_4 = 0.1$$

The matrix of pay-off between the decision and the results is indicated in the adjoining table :

	R_1	R_2	R_3	R_4
D_1	14	9	10	5
D_2	11	10	8	7
D_3	9	10	10	11
D_4	8	10	11	13

Show this decision situation in the form of a decision tree and indicate the most preferred decision and corresponding expected value.

Solution.

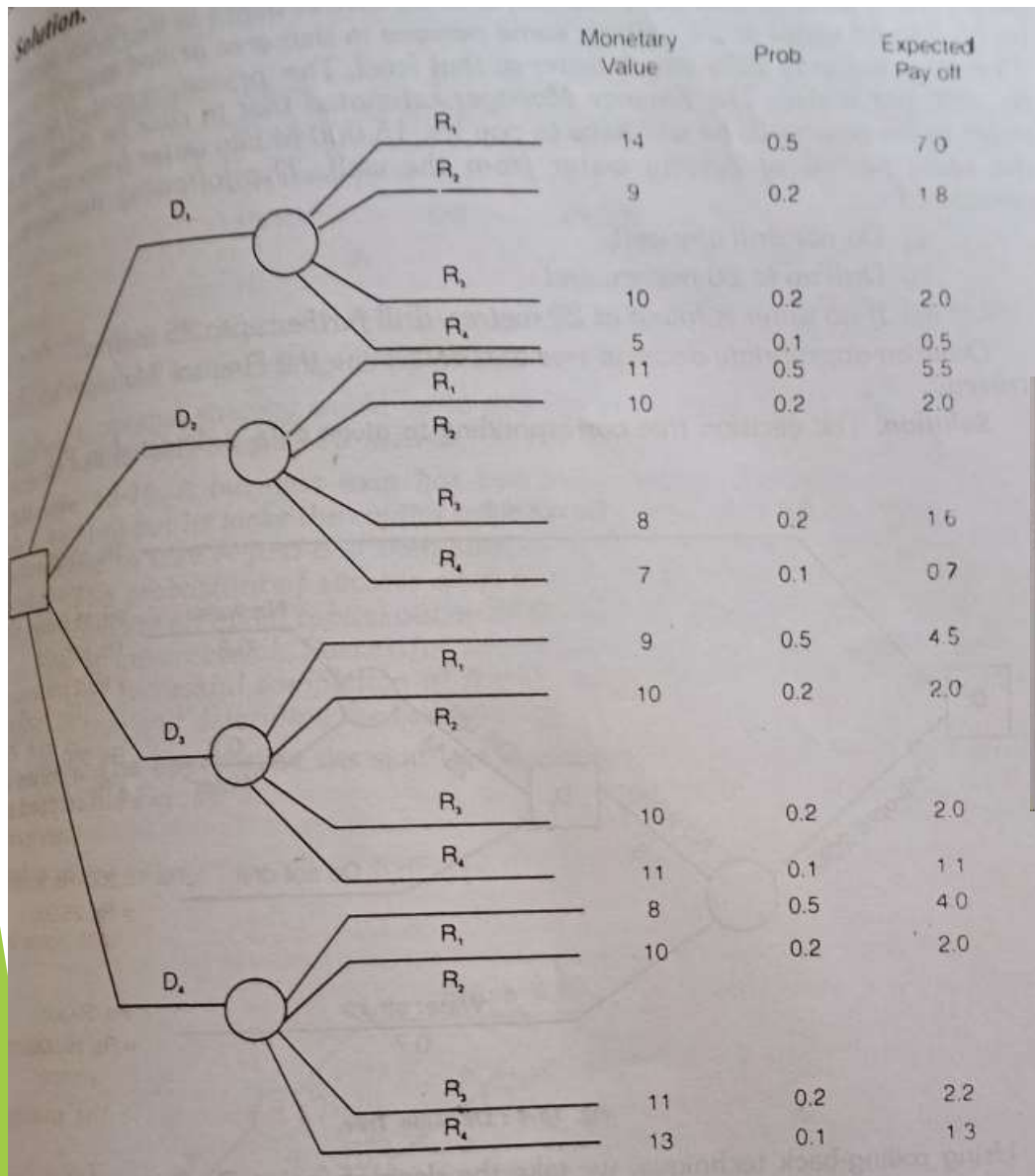


Fig. 10.3 : Decision Tree Diagram.

The most preferred decision at the decision node 1 is found by calculating expected value of each decision branch and selecting the path (course of action) with high value.

The expected monetary value for nodes A, B, and D is calculated as follows :

$$\text{EMV (A)} = 0.5 \times 14 + 0.2 \times 9 + 0.2 \times 10 + 0.1 \times 5 = 11.3$$

$$\text{EMV (B)} = 0.5 \times 11 + 0.2 \times 10 + 0.2 \times 8 + 0.1 \times 7 = 9.6$$

$$\text{EMV (C)} = 0.5 \times 9 + 0.2 \times 10 + 0.2 \times 10 + 0.1 \times 11 = 9.5$$

$$\text{EMV (D)} = 0.5 \times 8 + 0.2 \times 10 + 0.2 \times 11 + 0.1 \times 13 = 9.5$$

Since node A has the highest EMV, the decision at node A will be to

10-40

Example 10-18. A Finance Manager is considering drilling a well. In the past, only 70% of wells drilled were successful at 20 metres depth in that area. Moreover, on finding no water at 20 metres, some persons in that area drilled in further up to 25 metres but only 20% struck water at that level. The prevailing cost of drilling is Rs. 500 per metre. The Finance Manager estimated that in case he does not get water in his own well, he will have to pay Rs. 15,000 to buy water from outside for the same period of getting water from the well. The following decisions are considered :

- (i) Do not drill any well,
- (ii) Drill up to 20 metres, and
- (iii) If no water is found at 20 metres, drill further upto 25 metres.

Draw on appropriate decision tree and determine the Finance Manager's optimal strategy.

According to given data is depicted in Fig. 10.4

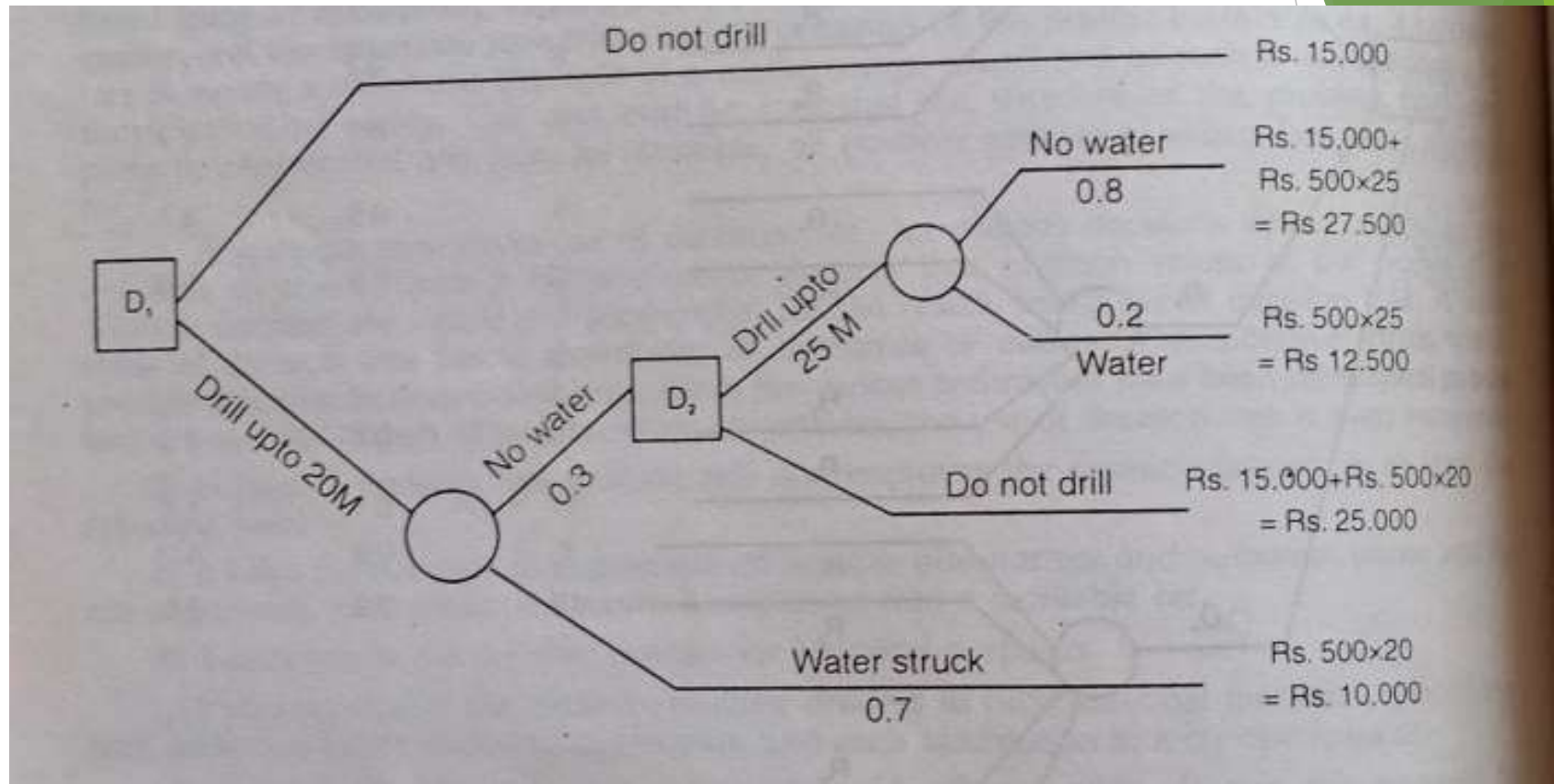


TABLE 10.42 : EVALUATION OF DECISION AND CHANCE NODES

Decision	Event	Prob.	Cash outflow (Rs.)	Expected cash outflow (Rs.)
Decision at point D_2				
1. Drill up to 25 metres	Water struck	0.2	12,500	2,500
	No water struck	0.8	27,500	22,000
				EMV (outflow) = 24,500
2. Do not drill				EMV (outflow) = 25,000

The decision at point D_2 is to drill up to 25 metres since the expected cash outflow for this decision is minimum, i.e., Rs. 24,500.

Decision at point D_1				
1. Drill up to 25 metres	Water struck	0.7	10,000	7,000
	No water struck	0.3	24,500	7,350
				EMV (outflow) = 14,350
2. Do not drill				EMV (outflow) = 15,000

The decision will be to drill up to 20 metres

Thus, the optimal strategy would be to drill up to 20 metres, and if no water is struck then further drill it up to 25 metres.

A glass factory specializing in crystal is experiencing a substantial backlog, and the firm's management is considering three courses of action:

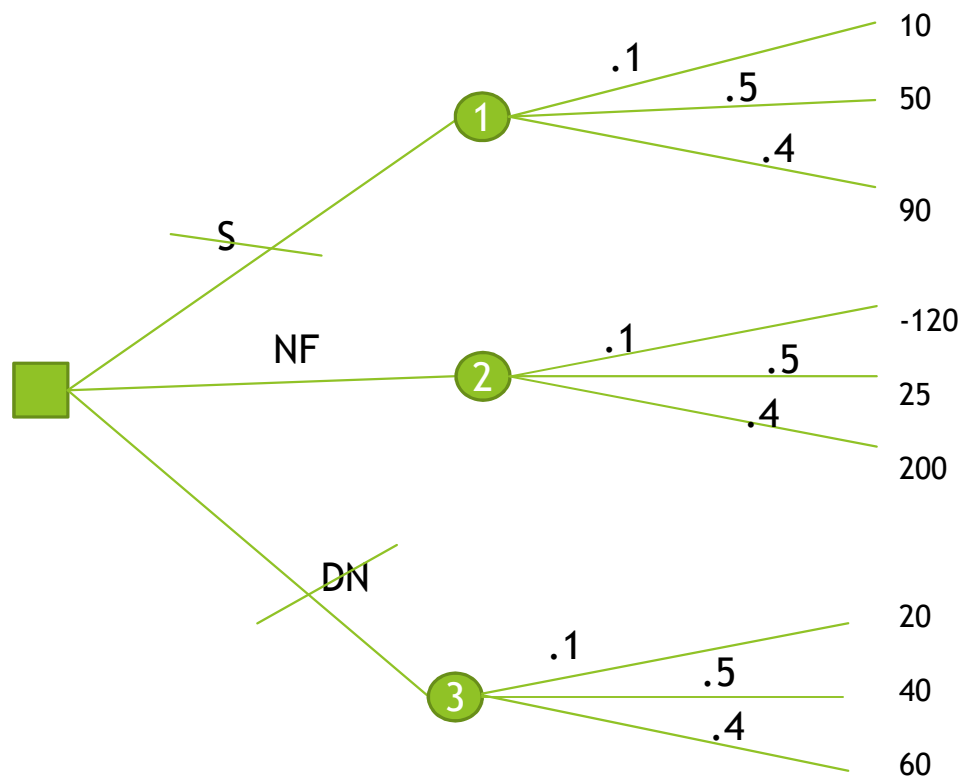
- A) Arrange for subcontracting
- B) Construct new facilities
- C) Do nothing (no change)

The correct choice depends largely upon demand, which may be low, medium, or high. By consensus, management estimates the respective demand probabilities as 0.1, 0.5, and 0.4.

The management estimates the profits when choosing from the three alternatives (A, B, and C) under the differing probable levels of demand. These profits, in thousands of dollars are presented in the table

manually create and solve this problem using a

	0.1	0.5	0.4
	Low	Medium	High
A	10	50	90
B	-120	25	200
C	20	40	60



$$ER1 = 10 \cdot 0.1 + 0.5 \cdot 50 + 0.4 \cdot 90 = 62$$

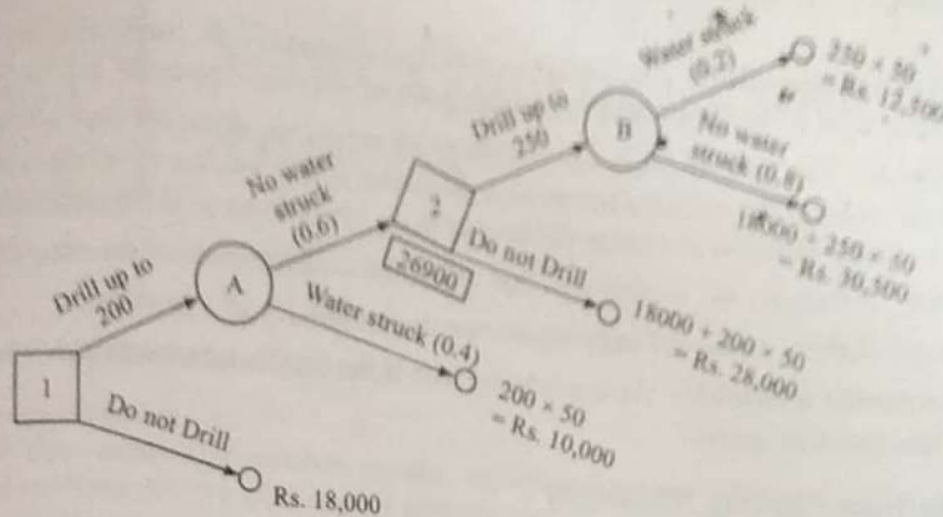
$$ER2 = -120 \cdot 0.1 + 0.5 \cdot 25 + 0.4 \cdot 200 = 80.5$$

$$ER3 = 20 \cdot 0.1 + 0.5 \cdot 40 + 0.4 \cdot 60 = 46$$

Example 21.12 A finance manager has to decide whether or not to drill a well on his farm. In his village, only 40 per cent of the wells drilled were successful at 200 feet of depth. Some of the farmers who did not get water at 200 feet, drilled further upto 250 feet but only 20 per cent struck water. Cost of drilling is Rs. 50 per foot. The finance manager estimated that he would pay Rs. 18,000 during a five year period on the present value terms, if he continues to buy water from the neighbour rather than go for the well, which would have a life of five years. He has three decisions to choose from: (a) Should he drill up to 200 feet, (b) If no water is found at 200 feet, should he drill up to 250 feet? (c) Should he continue to buy water from his neighbour?

Solution

1. Draw the decisions tree diagram for the given problem.



2. Compute the associated cost of each outcome and enter these values on the decision tree.

$$\begin{aligned}\text{EMV of node } B &= 0.2 \times 12500 + 0.8 \times 30500 \\ &= \text{Rs. } 26,900.\end{aligned}$$

$$\begin{aligned}\text{EMV of node } 2 &= \text{Min} [26900, 28000] \\ &= \text{Rs. } 26,900.\end{aligned}$$

$$\begin{aligned}\text{EMV of node } A &= 0.4 \times 10000 + 0.6 \times 26900 \\ &= \text{Rs. } 20,140.\end{aligned}$$

$$\begin{aligned}\text{EMV of node } 1 &= \text{Min} [20140, 18000] \\ &= \text{Rs. } 18,000.\end{aligned}$$

Result Thus the optimal (least) course of action for the manager is not to drill the well and pay Rs. 18,000 for water to his neighbour for five years.